

L5 Object Detection and Segmentation Quiz

ANS

1. What does an object detector predict for each detected object?

- A) A class label and a bounding box
- B) Only a class label
- C) Only a pixel-level mask
- D) A complete 3D model

ANS: A) A class label and a bounding box

2. Which four values are used to represent a bounding box in the slides?

- A) Class, score, depth, color
- B) x, y, width, height
- C) Row, column, class, loss
- D) Latitude, longitude, speed, heading

ANS: B) x, y, width, height

3. What does Intersection over Union (IoU) measure?

- A) The detector's inference time
- B) The number of object classes
- C) The overlap between a predicted box and a ground-truth box
- D) The image's pixel resolution

ANS: C) The overlap between a predicted box and a ground-truth box

4. What is the main purpose of Non-Maximum Suppression?

- A) To enlarge every predicted box
- B) To generate ground-truth labels
- C) To increase the number of duplicate detections
- D) To suppress lower-scoring overlapping boxes

ANS: D) To suppress lower-scoring overlapping boxes

5. What does precision measure for an object detector?

- A) The fraction of predicted positives that are correct
- B) The fraction of all actual positives that are detected
- C) The average image resolution
- D) The number of proposals generated per second

ANS: A) The fraction of predicted positives that are correct

6. What does recall measure for an object detector?

- A) The fraction of negative images classified correctly
- B) The fraction of actual positives that are detected
- C) The overlap of two bounding boxes
- D) The number of classes in the dataset

ANS: B) The fraction of actual positives that are detected

7. How does a sliding-window detector search an image?

- A) It labels all pixels simultaneously
- B) It generates only one fixed bounding box
- C) It moves a window across the image and classifies each patch
- D) It uses no CNN

ANS: C) It moves a window across the image and classifies each patch

8. What distinguishes a two-stage detector from a one-stage detector?

- A) It performs segmentation before receiving an image
- B) It uses two different cameras
- C) It predicts only object classes
- D) It first generates region proposals and then classifies and refines them

ANS: D) It first generates region proposals and then classifies and refines them

9. What is the purpose of region proposals?

- A) To produce a small set of boxes likely to contain objects
- B) To convert all objects into background
- C) To replace bounding-box regression
- D) To calculate image color histograms only

ANS: A) To produce a small set of boxes likely to contain objects

10. At test time, approximately how many region proposals does the original R-CNN process per image?

- A) 20
- B) 2,000
- C) 20,000
- D) 2

ANS: B) 2,000

11. Why is the original R-CNN computationally expensive?

- A) It performs no feature extraction
- B) It uses only one proposal
- C) It runs each region independently through the CNN
- D) It cannot resize region proposals

ANS: C) It runs each region independently through the CNN

12. What is the key efficiency improvement of Fast R-CNN over R-CNN?

- A) It removes bounding-box regression
- B) It uses no region proposals
- C) It processes each pixel with a separate CNN
- D) It extracts feature maps from the whole image once and shares them among regions

ANS: D) It extracts feature maps from the whole image once and shares them among regions

13. What component replaces Selective Search in Faster R-CNN?

- A) A Region Proposal Network
- B) A SoftMax-only network
- C) A nearest-neighbor upsampler
- D) A semantic segmentation mask

ANS: A) A Region Proposal Network

14. Which Faster R-CNN loss decides whether an anchor box contains an object?

- A) Object regression loss
- B) RPN classification loss
- C) Pixel-wise cross-entropy loss
- D) Image reconstruction loss

ANS: B) RPN classification loss

15. Which detector directly predicts classes and bounding boxes without a separate proposal stage?

- A) R-CNN
- B) Fast R-CNN
- C) YOLO
- D) Faster R-CNN

ANS: C) YOLO

16. How does YOLO organize the input image for prediction?

- A) As a single region only
- B) As an unordered list of pixels
- C) As approximately 2,000 Selective Search proposals
- D) As an n-by-n grid

ANS: D) As an n-by-n grid

17. In a YOLO grid-cell label, what does pc represent?

- A) The probability or indicator that an object is present
- B) The pixel color
- C) The proposal's computation time
- D) The number of object classes

ANS: A) The probability or indicator that an object is present

18. To which YOLO grid cell is an object assigned?

- A) Every cell overlapped by the object
- B) The cell containing the object's midpoint
- C) The upper-left cell only
- D) The cell with the smallest area

ANS: B) The cell containing the object's midpoint

19. For three classes, what is the length of the YOLO grid-cell vector containing pc, four box values, and class entries?

- A) 3
- B) 5

- C) 8
- D) 12

ANS: C) 8

20. What is the purpose of anchor boxes in YOLO?

- A) To remove all bounding-box predictions
- B) To force every object to have the same shape
- C) To perform semantic segmentation
- D) To provide predefined boxes with different scales and aspect ratios

ANS: D) To provide predefined boxes with different scales and aspect ratios

21. In the realistic YOLO example, what is the input image shape?

- A) 608 x 608 x 3
- B) 224 x 224 x 1
- C) 19 x 19 x 85
- D) 800 x 600 x 80

ANS: A) 608 x 608 x 3

22. In the YOLO output shape 19 x 19 x 5 x 85, what does 5 represent?

- A) The number of image channels
- B) The number of anchor boxes per grid cell
- C) The number of detection classes
- D) The number of backbone layers

ANS: B) The number of anchor boxes per grid cell

23. What does a Feature Pyramid Network produce?

- A) One class label for the entire image
- B) Only a full-resolution input copy
- C) Feature maps at multiple scales
- D) A list of ground-truth boxes

ANS: C) Feature maps at multiple scales

24. Which general performance tradeoff is described in the slides?

- A) Larger backbones are always faster
- B) One-stage detectors always have the best accuracy
- C) Two-stage detectors use no region proposals
- D) Two-stage detectors tend to be more accurate but slower

ANS: D) Two-stage detectors tend to be more accurate but slower

25. What does semantic segmentation assign to each pixel?

- A) A class label
- B) A separate object ID only
- C) A bounding box
- D) A region-proposal score

ANS: A) A class label

26. How does instance segmentation differ from semantic segmentation?

- A) It labels only background pixels
- B) It distinguishes separate instances of the same object class
- C) It returns only one label for the whole image
- D) It cannot identify object categories

ANS: B) It distinguishes separate instances of the same object class

27. Which category is considered 'stuff' rather than a countable 'thing'?

- A) Car
- B) Person
- C) Sky
- D) Cat

ANS: C) Sky

28. What characterizes a Fully Convolutional Network for semantic segmentation?

- A) It contains only fully connected layers
- B) It predicts only one box per image
- C) It requires Selective Search
- D) It uses convolutional layers without fully connected layers to predict all pixels

ANS: D) It uses convolutional layers without fully connected layers to predict all pixels

29. Why do efficient FCNs use downsampling followed by upsampling?

- A) To reduce computation and enlarge the receptive field, then restore output resolution
- B) To remove all spatial information permanently
- C) To convert segmentation into image classification
- D) To avoid using convolutional layers

ANS: A) To reduce computation and enlarge the receptive field, then restore output resolution

30. How is Mask R-CNN formed for instance segmentation?

- A) By removing the RPN from Faster R-CNN
- B) By adding a mask-prediction head to Faster R-CNN
- C) By replacing all convolution layers with SVMs
- D) By applying only nearest-neighbor interpolation

ANS: B) By adding a mask-prediction head to Faster R-CNN